

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE NAME: Fundamentals of Data Science COURSE CODE: DSA04

COURSE OUTCOME COVERED

***CO1:** Analyse the role of data science, facets of data, and the big data ecosystem for informed decision-making. (BL4)*

QUESTIONS: 05 Total Marks:100 Annexure A

CONCEPT MAPPING

Code of Conduct:

*I M.Gayathri(192472074)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:M.Gayathri

Annexure A

Concept Mapping

1. A college wants to analyze student performance using attendance records, internal marks,

assignment scores, and final exam results. Draw a concept map showing the Data Science process from data collection to data transformation.

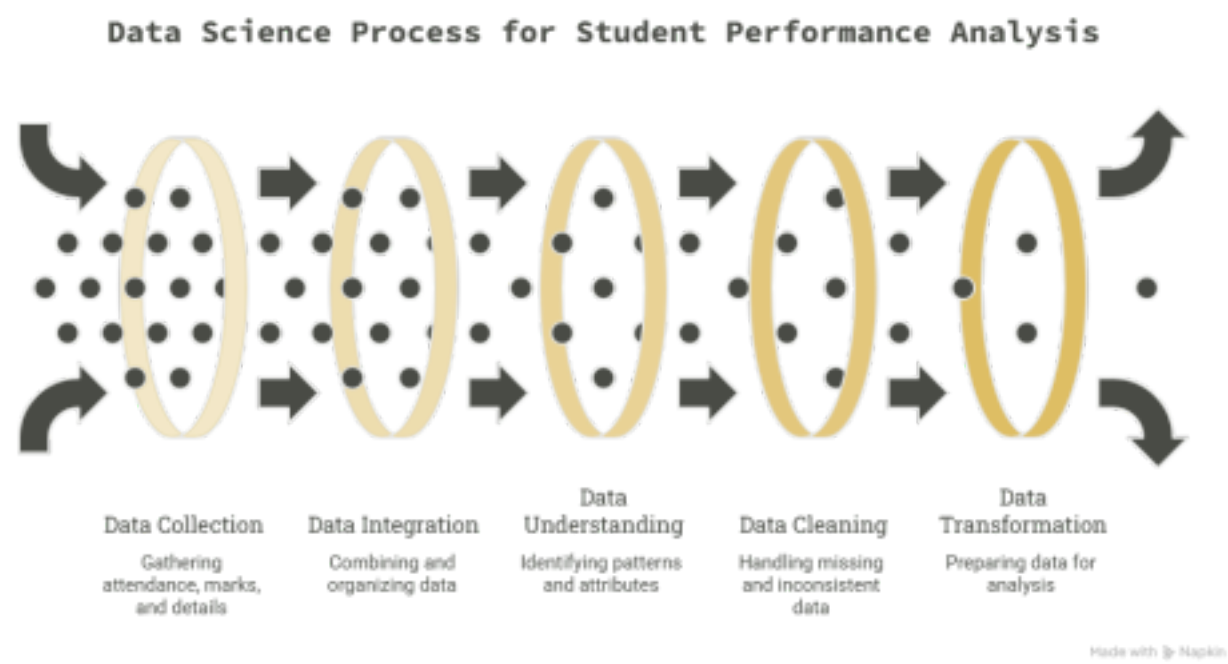
2. A hospital collects patient details from registration forms, lab reports, medical devices, and doctor notes. Construct a concept map showing the different data collection sources and how they are used for analysis.
3. An online shopping company finds missing customer details, duplicate orders, wrong product entries, and unusual purchase amounts in its dataset. Draw a concept map showing the data cleaning steps required before analysis.
4. A smart city project collects traffic data from CCTV cameras, GPS devices, sensors, and mobile applications. Prepare a concept map showing how data from multiple sources can be integrated into a single dataset.
5. A bank wants to prepare customer data for loan approval prediction. The dataset contains income, age, job type, loan amount, and credit score. Draw a concept map showing suitable data transformation techniques such as encoding, normalization, and feature scaling.

RUBRICS FOR CALCULATION

Concept Mapping			
Criteria	Weightage	Level 4 (Exemplary) Level 3 (Proficient) Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts Identifies most key from literature accurately concepts with minor and comprehensively omissions irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, Shows correct meaningful, and accurate relationships with relationships between minor gaps concepts Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured Mostly organized with minor hierarchically with logical structural issues flow and grouping Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to Shows some integration of Limited integration; mostly isolated	No integration of literature evidence

		show connections and literature concepts comparisons	
Clarity	10%	Concept map is clear, Generally clear with Clarity is reduced; neat, visually structured, minor readability difficult to interpret in and easy to interpret issues parts	Poorly presented and hardto understand

1.



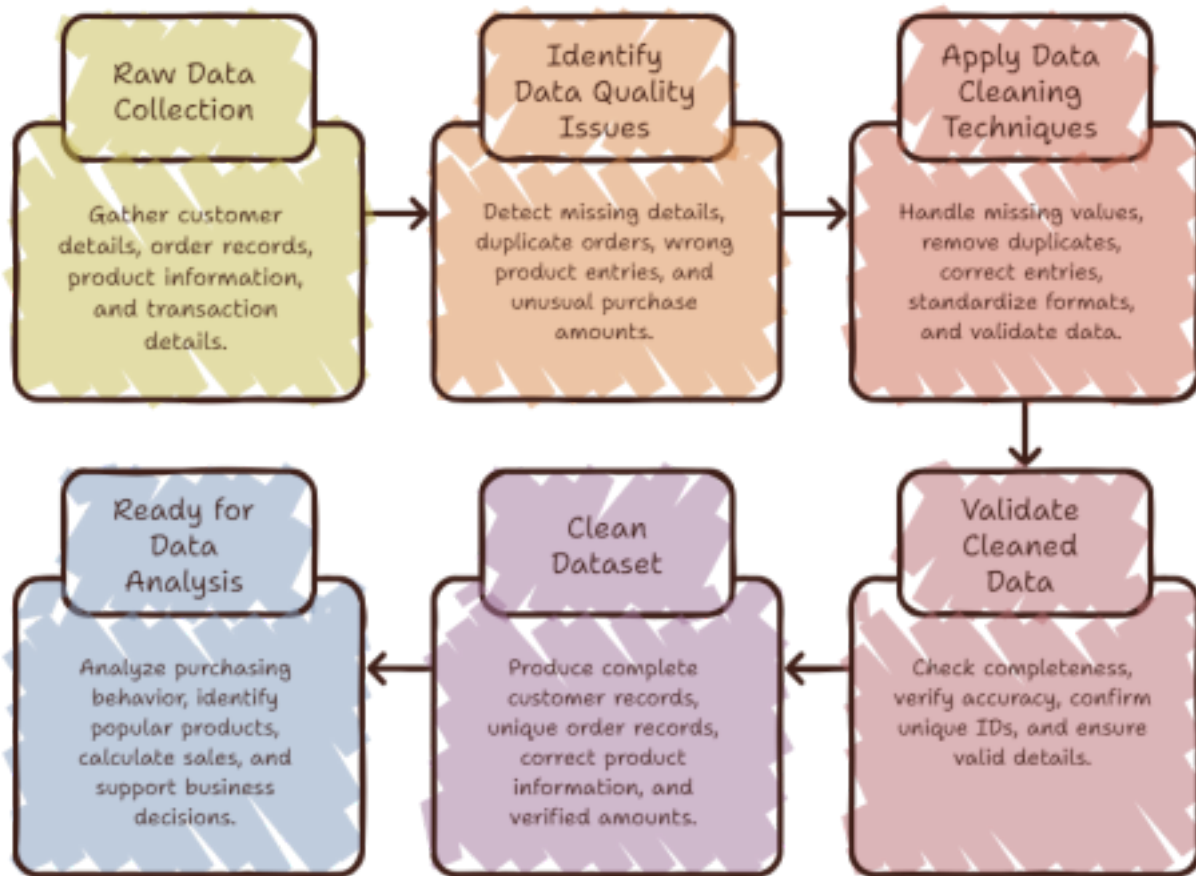
2.

Hospital Data Collection and Analysis Process



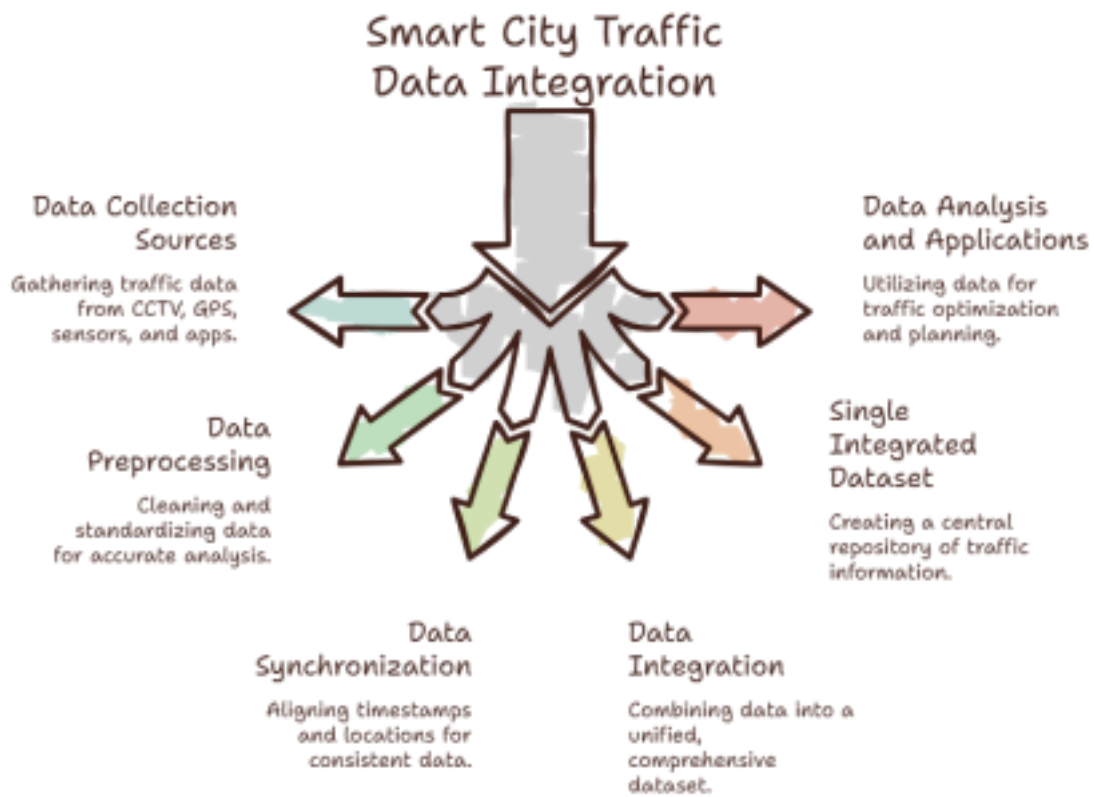
3.

Data Cleaning Process



Made with  Napkin

Unveiling Smart City Traffic Data Integration



Made with Napkin

Data Transformation for Loan Approval Prediction

